

MachineKeySessionSecurityTokenHandler class (System.IdentityModel.Services.Tokens)

MachineKeySessionSecurityTokenHandler 

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MachineKeySessionSecurityTokenHandler Class

Definition

Namespace: [System.IdentityModel.Services.Tokens](#)

Assembly: System.IdentityModel.Services.dll

Important

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Processes session tokens by using the signing and encryption keys specified in the ASP.NET `<machineKey>` element in the configuration file.

```
public ref class MachineKeySessionSecurityTokenHandler :  
    System::IdentityModel::Tokens::SessionSecurityTokenHandler
```

```
public class MachineKeySessionSecurityTokenHandler :  
    System.IdentityModel.Tokens.SessionSecurityTokenHandler
```

```
type MachineKeySessionSecurityTokenHandler = class  
    inherit SessionSecurityTokenHandler
```

```
Public Class MachineKeySessionSecurityTokenHandler  
    Inherits SessionSecurityTokenHandler
```

Inheritance

[Object](#)

[SecurityTokenHandler](#)

[SessionSecurityTokenHandler](#)

[MachineKeySessionSecurityTokenHandler](#)

Examples

The following XML shows how to explicitly specify the signing and encryption keys in configuration by using the ASP.NET `<machineKey>` element. The `<machineKey>` element is specified under the `<system.web>` element in the configuration file.

```
<machineKey compatibilityMode="Framework45" decryptionKey="CC510D ... 8925E6"  
validationKey="BEAC8 ... 6A4B1DE" />
```

The following XML shows how to add the [MachineKeySessionSecurityTokenHandler](#) to the token handler collection. The default [SessionSecurityTokenHandler](#) is first removed from the collection. Token handlers are configured under the `<securityTokenHandlers>` element.

```
<securityTokenHandlers>  
    <remove type="System.IdentityModel.Tokens.SessionSecurityTokenHandler,  
System.IdentityModel, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089"  
/>  
    <add type="System.IdentityModel.Services.Tokens.MachineKeySessionSecurityTokenHandler,  
System.IdentityModel.Services, Version=4.0.0.0, Culture=neutral,  
PublicKeyToken=b77a5c561934e089" />  
</securityTokenHandlers>
```

Remarks

By default, the [SessionSecurityTokenHandler](#) class uses the [ProtectedDataCookieTransform](#) class, which uses the Data Protection API (DPAPI), to protect session tokens. DPAPI provides protection by using user or computer credentials and stores the key data in the user profile. This means that a signed and encrypted session token cannot be validated or decrypted on another computer.

In contrast, the [MachineKeySessionSecurityTokenHandler](#) class uses the [MachineKeyTransform](#) class, which uses the cryptographic material specified in the `<machineKey>` element in the configuration file to protect the session cookie data. This means that the same keys (and session tokens) can be used on multiple computers. This is especially important when the application is deployed in a Web farm. For more information about how to protect applications deployed in a Web farm with Windows Identity Foundation, see [WIF and Web Farms](#).

Configure an application to use the [MachineKeySessionSecurityTokenHandler](#) by adding it to the token handler collection. If such a handler is present, you must first remove the [SessionSecurityTokenHandler](#) (or any handler that derives from the [SessionSecurityTokenHandler](#) class) from the token handler collection. This is because [MachineKeySessionSecurityTokenHandler](#) derives from [SessionSecurityTokenHandler](#) and a token handler collection cannot contain more than one handler of any given type.

Constructors

| Name | Description | | | [MachineKeySessionSecurityTokenHandler\(\)](#) |

Initializes a new instance of the [MachineKeySessionSecurityTokenHandler](#) class.

| | | [MachineKeySessionSecurityTokenHandler\(TimeSpan\)](#) |

Initializes a new instance of the [MachineKeySessionSecurityTokenHandler](#) class that has the specified default token lifetime.

| |

Properties

| Name | Description | | | [CanValidateToken](#) |

Gets a value that indicates whether this handler supports validation of tokens of the [SessionSecurityToken](#) type.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [CanWriteToken](#) |

Gets a value that indicates whether this handler can write tokens of the [SessionSecurityToken](#) type.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [Configuration](#) |

Gets or sets the [SecurityTokenHandlerConfiguration](#) object that provides configuration for the current instance.

(Inherited from [SecurityTokenHandler](#)) | | | [ContainingCollection](#) |

Gets the token handler collection that contains the current instance.

(Inherited from [SecurityTokenHandler](#)) | | | [CookieElementName](#) |

Gets the name of the cookie element.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [CookieNamespace](#) |

Gets the namespace of the cookie element.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [TokenLifetime](#) |

Gets or sets the token lifetime.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [TokenType](#) |

Gets the type of the tokens that this handler processes.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [Transforms](#) |

Gets the transforms that will be applied to the cookie.

(Inherited from [SessionSecurityTokenHandler](#)) | |

Methods

| Name | Description | | | [\[ApplyTransforms\(Byte\[\], Boolean\)\]](#)

([https://learn.microsoft.com/zh-](https://learn.microsoft.com/zh-cn/dotnet/api/system.identitymodel.tokens.sessionsecuritytokenhandler.applytransforms?view=netframework-4.8.1#system-identitymodel-tokens-sessionsecuritytokenhandler-applytransforms(system-byte()-system-boolean)))

[cn/dotnet/api/system.identitymodel.tokens.sessionsecuritytokenhandler.applytransforms?view=netframework-4.8.1#system-identitymodel-tokens-sessionsecuritytokenhandler-applytransforms\(system-byte\(\)-system-boolean\)\)](https://learn.microsoft.com/zh-cn/dotnet/api/system.identitymodel.tokens.sessionsecuritytokenhandler.applytransforms?view=netframework-4.8.1#system-identitymodel-tokens-sessionsecuritytokenhandler-applytransforms(system-byte()-system-boolean))) |

Applies the transforms specified by the [Transforms](#) property to encode or decode the specified cookie.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [CanReadKeyIdentifierClause\(XmlReader\)](#) |

Returns a value that indicates whether the XML element referred to by the specified XML reader is a key identifier clause that can be deserialized by this instance.

(Inherited from [SecurityTokenHandler](#)) | | | [CanReadToken\(String\)](#) |

Returns a value that indicates whether the specified string can be deserialized as a token of the type processed by this instance.

(Inherited from [SecurityTokenHandler](#)) | | | [CanReadToken\(XmlReader\)](#) |

Returns a value that indicates whether the reader is positioned on a `<wsc:SecurityContextToken>` element.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [CanWriteKeyIdentifierClause\(SecurityKeyIdentifierClause\)](#) |

Returns a value that indicates whether the specified key identifier clause can be serialized by this instance.

(Inherited from [SecurityTokenHandler](#)) | | | [CreateSecurityTokenReference\(SecurityToken, Boolean\)](#) |

When overridden in a derived class, creates a security token reference for tokens processed by that class. This method is typically called by a security token service (STS).

(Inherited from [SecurityTokenHandler](#)) | | | [CreateSessionSecurityToken\(ClaimsPrincipal, String, String, DateTime, DateTime\)](#) |

Creates a [SessionSecurityToken](#) based on the specified claims principal and the time range in which the token is valid.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [CreateToken\(SecurityTokenDescriptor\)](#) |

Creates a security token based on the specified token descriptor.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [DetectReplayedToken\(SecurityToken\)](#) |

When overridden in a derived class, throws an exception if the specified token is detected as being replayed.

(Inherited from [SecurityTokenHandler](#)) | | | [Equals\(Object\)](#) |

Determines whether the specified object is equal to the current object.

(Inherited from [Object](#)) | | | [GetHashCode\(\)](#) |

Serves as the default hash function.

(Inherited from [Object](#)) | | | [GetTokenTypeIdentifiers\(\)](#) |

Gets the token type URIs for the token types that this handler can process.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [GetType\(\)](#) |

Gets the [Type](#) of the current instance.

(Inherited from [Object](#)) | | | [LoadCustomConfiguration\(XmlNodeList\)](#) |

Loads custom configuration from XML.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [MemberwiseClone\(\)](#) |

Creates a shallow copy of the current [Object](#).

(Inherited from [Object](#)) | | | [ReadKeyIdentifierClause\(XmlReader\)](#) |

When overridden in a derived class, deserializes the XML referred to by the specified XML reader to a key identifier clause that references a token processed by the derived class.

(Inherited from [SecurityTokenHandler](#)) | | | [\[ReadToken\(Byte\[\], SecurityTokenResolver\)\]](#)

([https://learn.microsoft.com/zh-](https://learn.microsoft.com/zh-cn/dotnet/api/system.identitymodel.tokens.sessionsecuritytokenhandler.readtoken?view=netframework-4.8.1#system-identitymodel-tokens-sessionsecuritytokenhandler-readtoken(system-byte()-system-identitymodel-selectors-securitytokenresolver))

[cn/dotnet/api/system.identitymodel.tokens.sessionsecuritytokenhandler.readtoken?view=netframework-4.8.1#system-identitymodel-tokens-sessionsecuritytokenhandler-readtoken\(system-byte\(\)-system-identitymodel-selectors-securitytokenresolver\)](https://learn.microsoft.com/zh-cn/dotnet/api/system.identitymodel.tokens.sessionsecuritytokenhandler.readtoken?view=netframework-4.8.1#system-identitymodel-tokens-sessionsecuritytokenhandler-readtoken(system-byte()-system-identitymodel-selectors-securitytokenresolver))) |

Reads a [SessionSecurityToken](#) from a byte stream by using the specified token resolver.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [ReadToken\(String\)](#)) |

When overridden in a derived class, deserializes the specified string into a token of the type processed by the derived class.

(Inherited from [SecurityTokenHandler](#)) | | | [ReadToken\(XmlReader, SecurityTokenResolver\)](#)) |

Reads a [SessionSecurityToken](#) by using the specified XML reader and token resolver.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [ReadToken\(XmlReader\)](#)) |

Reads a [SessionSecurityToken](#) by using the specified XML reader.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [SetTransforms\(IEnumerable<CookieTransform>\)\)](#)) |

Sets the transforms that will be applied to the cookie.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [ToString\(\)](#) |

Returns a string that represents the current object.

(Inherited from [Object](#)) | | | [TraceTokenValidationFailure\(SecurityToken, String\)](#)) |

Traces a failure event during security token validation when tracing is enabled.

(Inherited from [SecurityTokenHandler](#)) | | | [TraceTokenValidationSuccess\(SecurityToken\)](#)) |

Traces a successful security token validation event when tracing is enabled.

(Inherited from [SecurityTokenHandler](#)) | | | [ValidateSession\(SessionSecurityToken\)](#)) |

Determines whether the session associated with the specified token is still valid. Validity is determined by checking the [ValidFrom](#) and [ValidTo](#) properties of the specified token. If the session is no longer valid, an exception is thrown.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [ValidateToken\(SecurityToken\)](#)) |

Validates the specified token and returns its claims.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [ValidateToken\(SessionSecurityToken, String\)](#)) |

Validates the specified session token and returns its claims.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [WriteKeyIdentifierClause\(XmlWriter, SecurityKeyIdentifierClause\)](#)) |

When overridden in a derived class, serializes the specified key identifier clause to XML. The key identifier clause must be of a type supported by the derived class.

(Inherited from [SecurityTokenHandler](#)) | | | [WriteToken\(SecurityToken\)](#)) |

When overridden in a derived class, serializes the specified security token to a string. The token must be of a type processed by the derived class.

(Inherited from [SecurityTokenHandler](#)) | | | [WriteToken\(SessionSecurityToken\)](#)) |

Serializes the specified token to a byte array.

(Inherited from [SessionSecurityTokenHandler](#)) | | | [WriteToken\(XmlWriter, SecurityToken\)](#)) |

Serializes the specified token by using the specified XML writer.

(Inherited from [SessionSecurityTokenHandler](#)) | |

Applies to

See also

- [SessionSecurityTokenHandler](#)
- [WIF and Web Farms](#)

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